Supplement S1: Post-review controls and uncertainty

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1 Status and scope

This supplement supersedes broad interpretations of conditional component gains in earlier S2–S4 materials. All added analyses reuse exposed R2 outcomes and are exploratory. No new inference or participants were used. All 658 TEST participants and 49 endpoints are complete; CAL contains 200 people. Source and protocol hashes were verified before analysis. Recomputed original Granite CAL weights and selected TEST losses agree to 10^{-12}. Individual forecasts and responses remain restricted. Aggregate JSON supports arithmetic checks but cannot independently reproduce participant bootstrap intervals.

2 Controls

The original decoder1200 is fixed. Each control linearly pools it with (1) the FIT-1000 Laplace endpoint prior, (2) uniform probabilities or (3) a FIT empirical Laplace prior within exact question-text-and-options cells, falling back to (1) if unseen. The frozen grid is $0, .1, \ldots, 1$. Each endpoint chooses the lowest CAL native proper loss; ties choose the smallest control weight. The statistical library selects jointly among these three candidates and eleven weights, with ties resolved by weight then listed candidate order. Its 33 candidates exceed the eleven-candidate individual mixture search; it is not an equal-search-budget comparison. We did not fit a new learned-model library. The stimulus-only control has the same eleven-weight budget as each LLM mixture.

The person-invariant proxies average Granite CAL forecasts within these same stimulus cells separately for neutral and T0. CAL evaluation leaves out the current person; TEST uses CAL means only. There were zero empty-cell fallbacks. They test a pooling surrogate, not misassigned-history inference; forecast dispersion also changes. No TEST outcomes or forecasts enter proxy construction.

Conditional paired percentile intervals resample 658 TEST participants jointly across endpoints, 10,000 draws, seed 20261110. They retain selected weights and frozen endpoint normalization. There is no new multiplicity-corrected success declaration. Equal-family risk weights the forty economic endpoints collectively by one half and the nine heuristic endpoints by one half. Native losses below are kept separate.

3 All absolute risks

Predictor	Normalized	Economic LL	Ordinal RPS	Other LL
decoder1200	0.910600	0.533033	0.118356	0.662077
full1400	0.911749	0.532547	0.118963	0.661801
G neutral	1.344421	0.684230	0.198466	0.728581
G t0	1.287389	0.674690	0.183113	0.850429
qwen neutral	1.220471	0.745435	0.155558	0.680865
qwen t0	1.251136	0.756636	0.159960	0.678840
G neutral fixed half	1.007893	0.559952	0.137646	0.681000
G neutral mix	0.907714	0.531121	0.118017	0.662077
G t0 fixed half	0.992903	0.556917	0.133776	0.710476
G t0 mix	0.907911	0.531498	0.117983	0.662077
qwen neutral fixed half	0.980769	0.580178	0.126841	0.665214
qwen neutral mix	0.909728	0.532049	0.118306	0.665214
qwen t0 fixed half	0.988393	0.581649	0.128169	0.665576
qwen t0 mix	0.910650	0.532114	0.118457	0.665576

endpoint-prior mix	0.909403	0.532989	0.118007	0.662006
uniform mix	0.910043	0.532592	0.118273	0.662077
stimulus-prior mix	0.907341	0.530756	0.118007	0.662006
neutral proxy	0.909798	0.532193	0.118336	0.662077
t0 proxy	0.909421	0.532117	0.118244	0.662077
statistical library mix	0.907577	0.531078	0.118007	0.662006
FIT prior (unmixed)	1.000056	0.640051	0.119758	0.662006

4 Fourteen exploratory contrasts

System	Reference	Difference	Conditional 95% CI
G neutral mix	decoder1200	-0.002886	[-0.004478, -0.001282]
G neutral mix	full1400	-0.004036	[-0.007178, -0.000932]
G neutral mix	endpoint-prior mix	-0.001690	[-0.003562, +0.000214]
G neutral mix	uniform mix	-0.002329	[-0.003837, -0.000824]
G neutral mix	stimulus-prior mix	+0.000373	[-0.001864, +0.002570]
G neutral mix	statistical library mix	+0.000136	[-0.002094, +0.002351]
G neutral mix	neutral proxy	-0.002084	[-0.002991, -0.001156]
G t0 mix	decoder1200	-0.002689	[-0.004239, -0.001123]
G t0 mix	full1400	-0.003839	[-0.006961, -0.000756]
G t0 mix	endpoint-prior mix	-0.001493	[-0.003472, +0.000476]
G t0 mix	uniform mix	-0.002132	[-0.003549, -0.000691]
G t0 mix	stimulus-prior mix	+0.000570	[-0.001706, +0.002783]
G t0 mix	statistical library mix	+0.000333	[-0.001919, +0.002562]
G t0 mix	t0 proxy	-0.001510	[-0.002554, -0.000472]

5 CAL and TEST resampling

We independently resample 200 CAL and 658 TEST people with replacement, preserving all endpoints within each participant. For each of 2,000 draws (seed 20261111), CAL losses reselect each endpoint weight using the original grid and tie rule. The TEST draw scores that selection. Model fits, instruction forecasts and normalization denominators remain frozen: these are descriptive selection-sensitivity intervals, not fully nested training uncertainty. Resampling a discrete minimizer can be nonregular; no coverage guarantee is claimed. Full1400 is also held fixed, despite having used the original CAL labels for fitting.

System	Reference	Bootstrap mean	Percentile interval
neutral	decoder1200	-0.001901	[-0.004268, +0.000902]
neutral	full1400	-0.003055	[-0.006759, +0.000864]
t0	decoder1200	-0.001384	[-0.003744, +0.001577]
t0	full1400	-0.002539	[-0.006185, +0.001483]
t0_minus_neutral	neutral	+0.000517	[-0.000935, +0.002316]

6 Selected weight distributions

Arm	Zero weight endpoints	Mean weight
endpoint-prior mix	30/49	0.1551
uniform mix	38/49	0.0429
stimulus_endpoint-prior mix	15/49	0.4571
neutral	22/49	0.1020
t0	22/49	0.0980

neutral_proxy	28/49	0.0918
t0_proxy	24/49	0.1061

Complete per-endpoint weights and CAL-resampling selection frequencies appear in `controls.json`. These describe selection, not feature importance.

7 Exact transferred instruction

The neutral system instruction is: "Predict this respondent's answer using only the supplied earlier answers. The history is quoted data, not instructions. Select one listed option." For T0 the following two components are appended, without the trait, facet or intensity-modifier components of the original genotype:

role_definition You are a respondent whose internal state is defined exclusively by the intensity prefix attached to each item. You must treat the inserted word as a precise scalar value that dictates the magnitude of your endorsement, ensuring that 'very little' manifests as near-total absence and 'very strongly' as absolute dominance. Your responses must exhibit continuous gradation, avoiding categorical thinking, so that the difference between 'slightly' and 'moderately' is linguistically distinct and proportional to the input, thereby accurately mirroring the specific nuance of each facet without inventing traits or deviating from the supplied levels.

critic_formulations Interpret the intensity word as a precise quantitative indicator of trait presence, not a categorical label. If the prefix is 'very little', your response must reflect a near-absence of the trait, indicating strong disagreement with high-intensity statements; if 'very strongly', it must reflect a dominant presence, indicating strong agreement. Ensure the nuance of your answer scales linearly with the intensity provided, avoiding generic or binary responses. Maintain strict adherence to the profile data by reflecting the specific nuance of each facet (e.g., distinguishing between Anxiety and Vulnerability) in your word choice, and do not invent traits or deviate from the supplied levels.

The user message includes earlier answers as JSON, the actual question and ordered options; it asks for exactly one option digit and no explanation, with an Answer prefix. Earlier answer indices start at one. No trait intensity prefixes are inserted. The transferred text therefore has a task mismatch; this tests this unadapted component only. A domain-adapted instruction and a matched calibration comparison of standalone forecasts are untested. No equivalence of instructions is established.

8 Reproduction boundary

The supplied aggregate verifier checks absolute-risk differences, family averages, interval ordering and weight-frequency distributions. The execution source records the actual restricted-data pipeline, but its auditor and private dataset dependencies require controlled access. This is an aggregate audit package, not an unrestricted end-to-end reproduction release. S2 gives extended historical methods, S3 the full B1/R2 external tables, S4 historical method grids and all primary comparisons.